

CS-E4780 Scalable Systems and Data Management

Course Project Evaluation Criteria (2026)

Detecting Trading Trends in Financial Tick Data

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The course project is graded out of **26 points**: system implementation (12), final report (12), and teamwork and individual contributions (2). Each criterion in the implementation and report tables is scored on a four-level scale from Unacceptable (0) to Exceptional (3). The criteria refer to the queries and requirements defined in the project description: Query 1 (exponential moving averages over 5-minute tumbling windows), Query 2 (breakout detection producing buy/sell advisories), and the optional smart visualization. The total maps to course grades as shown on the right.

Grade	Points
5	24 – 26
4	19 – 23
3	15 – 18
2	11 – 14
1	1 – 10
0	0

System implementation (12 points)

Criterion	Unacceptable (0)	Marginal (1)	Acceptable (2)	Exceptional (3)
Functionality	Unable to correctly ingest or process the real tick data; neither query produces results.	Basic functionality: the data is ingested and EMAs (Query 1) are computed, but results contain frequent errors, or crossover detection (Query 2) is missing or unreliable.	Both queries work as intended with minor bugs, e.g. occasional incorrect EMA values, or missed or spurious buy/sell advisories.	Full functionality: correct EMA ₃₈ and EMA ₁₀₀ per window and correct buy/sell advisories for all subscribed symbols, with no significant issues.
Performance	System is unable to process even small portions of the dataset without significant lag.	Can process part of the dataset but struggles with larger portions or with real-time processing requirements.	Processes large portions of the dataset efficiently, with occasional performance dips.	Processes the entire week of data smoothly, sustaining high throughput with low and stable latency.
Scalability	Cannot scale to handle multiple exchanges or high event volumes.	Limited scalability; handles data from a single exchange or a reduced dataset only.	Scales to handle data across all three exchanges with moderate event volumes.	Scales effectively across all exchanges and event volumes, including peak rates and the long-tail symbol distribution, with capacity for further increases (more symbols, more subscribers, more resources).
Creative explorations	No exploration of the design space; a single approach with no justification.	Some alternatives are mentioned (e.g. windowing or state management choices), but trade-offs are not analysed.	Balanced trade-offs in the design space and system architecture are discussed and justified, e.g. partitioning by symbol, state management, or handling of late and out-of-order events.	Well-reasoned exploration of the design space supported by measurements, and/or a smart visualization that makes the query results directly interpretable (Section 3.4 of the project description).

Final report (12 points)

The report follows the ACM proceedings template (L^AT_EX or Word), is 4–5 pages long, and is submitted together with a link to the system’s GitHub repository.

Criterion	Unacceptable (0)	Marginal (1)	Acceptable (2)	Exceptional (3)
System design and architecture	No clear description of the system design.	Basic design, but lacks support for real-time or complex event processing needs.	Includes design elements for real-time data processing, metadata handling, and normalization.	Well-architected system design that handles complex event processing, real-time requirements, and scalability.
Implementation explanation	Explanation of the implementation is missing or very unclear.	Limited details provided, with minimal explanation of the handling of tick and housekeeping events.	Clear explanation of how tick, housekeeping, and metadata events are processed, including the hardware or cloud services and programming tools used.	Detailed explanation covering all implementation aspects, including event handling, windowing and EMA state, crossover detection, normalization, and metadata usage.
Evaluation and performance	No meaningful evaluation or performance testing presented.	Basic evaluation, with limited testing on data handling for events and metadata.	Performance testing covers correctness of the query results and handling of different event types and distribution patterns.	Comprehensive evaluation: correctness, throughput and latency under load, scalability across exchanges and with varying resources (CPU/GPU utilization), and handling of long-tail distributions.
Academic writing	Report is poorly structured and lacks coherence.	Basic structure, but lacks clarity in discussing data handling and event distribution.	Well-structured and clear, with minor issues in presentation or analysis. Citation format is consistent.	Professionally written, clear, and logically structured, with a strong focus on dataset analysis. Follows the template and page limit.

Teamwork and individual contributions (2 points)

Each team member must answer the following questions as *individual contributions* in the report:

1. What are the most important decisions you and your group have made about the system design?
2. What are the most challenging parts during the design and implementation?
3. What is your specific contribution in the team? How do you characterize the team’s overall functionality?

Criterion	Unacceptable (0)	Acceptable (1)
Individual contributions	No meaningful individual contributions documented.	Clear description of contributions and reflection.
Teamwork	Team collaboration is ineffective, and roles are unclear.	Exemplary teamwork, with clear roles, efficient data management, and active collaboration.

Students working alone are evaluated as a one-member team using the same criteria, except for the teamwork part.